

Temperature Limits and Safety Considerations for Gas Cylinders in a Heated Cabinet (A) - D2.036

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1 Introduction

This document summarises the temperature limits, pressure behaviour, and safety considerations for several compressed gas and refrigerant cylinders intended for storage inside a heated cabinet. The purpose of this report is to assist with engineering and safety planning for cabinet heating systems where gas cylinders may be exposed to elevated temperatures.

2 General Safety Guidance

For nearly all industrial gas and refrigerant cylinders, the commonly accepted maximum external cylinder temperature is:

$$T_{\max} \approx 50\text{--}52\text{ }^{\circ}\text{C}$$

This limit is widely used by industrial gas suppliers and refrigerant manufacturers. The reasons for this limit include:

- Rapid increase of internal pressure with temperature.
- Possible activation of pressure relief devices.
- Risk of cylinder rupture if severely overheated.
- Increased fire and explosion hazards for flammable gases.
- Increased toxicity risks from decomposition products in refrigerants.

2.1 Important Cabinet Design Recommendations

- Cabinet air temperature should normally remain below 45 °C to maintain engineering margin.
- Local heater hot-spots must not directly irradiate cylinder walls.
- Forced-air circulation is strongly recommended.
- Independent over-temperature protection should be installed.
- Continuous temperature monitoring is recommended.
- Ventilation is required for all flammable gases and refrigerants.

3 Cylinder Summary Table

Gas / Refrigerant	UN Number	Storage Type	Recommended Max Temp	Main Hazards
R134a	UN3159	Liquefied refrigerant	50–52 °C	Pressure rise, venting, HF decomposition products
SF ₆	UN1080	Compressed gas	50–52 °C	High pressure, asphyxiation, toxic arc decomposition products
Neon	UN1065	Compressed gas	50–52 °C	High stored energy, asphyxiation
Isobutane	UN1969	Liquefied flammable gas	50 °C	Fire, explosion, vent ignition
Krypton	UN1056	Compressed gas	50–52 °C	High stored energy, asphyxiation
R1234yf	UN3161	Liquefied refrigerant	50–52 °C	Mild flammability, HF decomposition
R1234ze	UN3163	Liquefied refrigerant	50–52 °C	Mild flammability, pressure rise
R1233zd(E)	UN3163	Low-pressure refrigerant	50–52 °C	Decomposition products, pressure rise

4 Detailed Gas Information

4.1 R134a (UN3159)

Chemical name:

1,1,1,2-Tetrafluoroethane

R134a is a non-flammable HFC refrigerant stored as a liquefied gas.

Approximate saturation pressures:

- 6 bar at 20 °C
- 13 bar at 50 °C

Primary risks:

- Rapid pressure increase during heating.
- Pressure relief valve activation.
- Toxic hydrogen fluoride generation during fire decomposition.

4.2 SF₆ (UN1080)

Sulfur hexafluoride is a dense inert insulating gas stored at high pressure.

Approximate pressures:

- 20.7 bar at 20 °C

- Approximately 35 bar to 40 bar near 50 °C

Primary risks:

- High-pressure stored energy.
- Oxygen displacement in confined spaces.
- Toxic decomposition products after electrical arcing.

4.3 Neon (UN1065)

Neon is an inert noble gas stored as a compressed gas.

Typical cylinder conditions:

- Fill pressure typically 200 bar to 230 bar
- Hydrostatic test pressure commonly above 300 bar

Estimated pressure increase:

- Approximately 200 bar at 15 °C
- Approximately 225 bar at 50 °C

Primary risks:

- Large stored mechanical energy.
- High-velocity gas release if damaged.
- Asphyxiation risk.

4.4 Isobutane (UN1969)

Isobutane is a liquefied flammable hydrocarbon gas.

Approximate vapour pressures:

- 3.1 bar at 20 °C
- 5.3 bar at 40 °C
- 6.9 bar at 50 °C

Primary risks:

- Fire and explosion.
- Flammable vent release.
- BLEVE-type failure during severe overheating.

Special precautions:

- Ventilation is mandatory.
- Ignition sources must be eliminated.
- Heater elements must be isolated from leaked gas.

4.5 Krypton (UN1056)

Krypton is an inert noble gas stored at high pressure.

Typical operating conditions:

- Typical fill pressure: 200 bar to 230 bar
- Typical hydrostatic test pressure: above 300 bar

Estimated pressure increase:

- Approximately 200 bar at 15 °C
- Approximately 225 bar at 50 °C

Primary risks:

- Stored mechanical energy.
- Asphyxiation hazard.

4.6 R1234yf (UN3161)

Chemical name:

2,3,3,3-Tetrafluoropropene

R1234yf is classified as an A2L mildly flammable refrigerant.

Approximate pressures:

- 6.7 bar at 20 °C
- 13 bar to 14 bar at 50 °C

Primary risks:

- Mild flammability.
- Pressure rise during heating.
- Hydrogen fluoride production in fire conditions.

4.7 R1234ze (UN3163)

R1234ze is an A2L mildly flammable low-pressure refrigerant.

Approximate pressures:

- 3.5 bar at 20 °C
- Approximately 7 bar at 50 °C

Primary risks:

- Mild flammability.
- Pressure increase during heating.

4.8 R1233zd(E) (UN3163)

Chemical name:

trans-1-chloro-3,3,3-trifluoropropene

R1233zd(E) is a low-pressure refrigerant classified as A1 (non-flammable under standard conditions).

Approximate pressures:

- Approximately 0.8 bar(g) at 20 °C
- Approximately 2 bar(g) at 50 °C

Primary risks:

- Toxic decomposition products during fire.
- Pressure rise if overheated.

5 Engineering Recommendations for Heated Cabinets

5.1 Recommended Maximum Cabinet Temperature

For mixed gas storage, a conservative cabinet air temperature limit is:

$$T_{cabinet} \leq 45 \text{ }^{\circ}\text{C}$$

This provides margin below the standard cylinder limit of approximately 50 °C.

5.2 Recommended Protection Features

- Dual independent temperature sensors.
- Over-temperature shutdown.
- Forced-air circulation fans.
- Thermal insulation between heaters and cylinders.
- Flammable gas detection for isobutane and A2L refrigerants.
- Ventilation or pressure relief vents.

5.3 Heater Design Considerations

The heater should:

- Avoid direct radiant heating of cylinders.
- Maintain uniform air temperature.
- Use low surface temperature elements where possible.
- Prevent local cylinder wall temperatures from exceeding safe limits.

6 Conclusion

All listed cylinders should be treated as pressure vessels with significant stored energy. Even inert gases become hazardous when overheated due to pressure rise.

For engineering safety:

- **Maintain cabinet temperatures below 45 °C.**
- **Never exceed cylinder wall temperatures of 50 °C to 52 °C.**
- Provide ventilation, monitoring, and over-temperature protection.
- Apply additional precautions for flammable gases and refrigerants.